

Movement Logger — Race Mode Setup

Live rider tracking: every rider's phone streams its GPS position to one shared map on the desktop.

Requirements

- All phones and the desktop computer on the same WiFi network (club/venue WiFi works fine).
- Versions: Desktop app 0.0.67 or newer, Android app 0.0.52 or newer, iOS app 1.0.29 or newer.
- Android riders: a u-blox USB GNSS receiver plugged into the phone. iPhone riders: nothing extra.

1 — Desktop (race committee)

1. Install: github.com/zdavatz/movement_logger_desktop -> Releases (macOS DMG, Windows zip, Linux tar).
 2. Open the app -> Race tab -> click "Start listening".
 3. The header now shows: Phones send to <IP>:47777 — riders enter exactly these two values.
 4. Map controls: scroll or double-click or the +/- buttons to zoom; "Satellite" switches to imagery; "Follow riders" keeps everyone in view automatically. macOS firewall prompt: click Allow.
- Every received fix is also logged to `~/movementlogger/race/race_<timestamp>.csv` for post-race analysis.

2 — Android rider (u-blox USB GPS)

1. Install "Movement Logger" from Google Play.
2. Plug the u-blox receiver into USB-C -> GPS tab -> Connect (allow USB access).
3. In the "Race mode" card: enter your rider name + the desktop IP and port from step 1.3 -> toggle ON.
4. "Sending — N fixes" counts up as soon as the receiver has a GPS fix (needs sky view — go outside).

3 — iPhone / Apple Watch rider

1. Install "Movement Logger" from the App Store.
2. GPS tab -> "Race mode" card: rider name + the desktop IP and port, then pick the GPS source:
 - iPhone GPS: toggle ON — done. The phone's own GPS starts automatically.
 - Apple Watch: toggle ON, then start a recording in the watch app. The watch streams its fixes through the paired iPhone — keep the iPhone with you (it is the uplink).

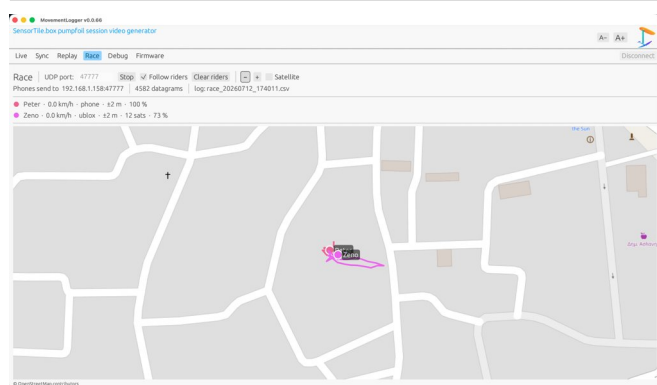
Reading the map

- Dot = the rider's latest position; the translucent circle around it is the receiver's own accuracy estimate (bigger circle = less precise fix). The coloured trail is the recently travelled path.
- The rider list shows: name, speed, source (ublox / phone / watch), +-accuracy, satellites, battery.
- A rider greyed out has sent nothing for 5 s (out of range, capsized, app closed) — the last position stays on the map on purpose.

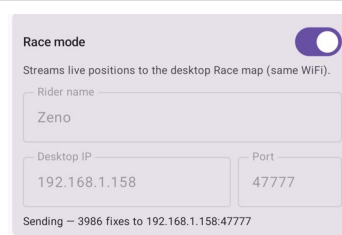
Troubleshooting

- No dot appears: same WiFi? IP and port typed exactly as the desktop shows? GPS fix yet (sky view)?
- Everything jumps or looks imprecise near buildings: watch the accuracy circle — GPS is a 2-5 m instrument; on open water it is at its best.
- Two riders must not use the same rider name on the same source type.

What it looks like



Desktop — Race tab



Android — Race mode card



iPhone — Race mode card